

Project Proposal

Retrieval-Augmented Audio Captioning: RECAP

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Project Summary

Audio captioning is essentially the task of generating a natural language description of the contents of an audio clip. It is a sequence-to-sequence problem mapping variable-length audio waveforms to variable-length text. The core challenge is multimodal: the model must learn to ground language in acoustic features. A secondary challenge, and the motivation for applying retrieval-augmented generation (RAG), is domain shift. Models trained on one audio captioning dataset often fail to describe audio events they have not encountered during training. We study RECAP ([Ghosh et al., 2024](#)), which addresses this by conditioning caption generation on similar captions retrieved from an external datastore. This concept can be applied for real-world applications like video content moderation or an app for audio journaling.

Approach

We will implement the RECAP architecture from scratch in PyTorch, following the design described by [\[Ghosh et al., 2024\]](#). The architecture consists of three main components: (1) a frozen CLAP audio encoder [\[Wu et al., 2023\]](#) that extracts last-hidden-state representations from the penultimate layer, (2) a frozen GPT-2 language decoder, and (3) learnable cross-attention layers inserted at each GPT-2 decoder layer to condition the language model on the audio embedding sequence. For retrieval-augmented prompt construction, we will use the CLAP model to compute cosine similarity between the input audio embedding and all text caption embeddings in a datastore, retrieve the top-k ($k=4$) most similar captions, and format them into a structured prompt of the form: "Audios similar to this audio sounds like: [caption 1], [caption 2], ... [caption k]. This audio sounds like:". We plan to use the official CLAP checkpoint from LAION and the Hugging Face GPT-2 base model as our frozen backbones.

Beyond reproduction, we plan to conduct several analytical experiments. First, we will perform an ablation study comparing performance with and without retrieved prompts, and with different values of k , to quantify the contribution of the retrieval component. Second, we will analyze the effect of datastore composition and size on captioning quality, comparing training-set-only datastores against augmented datastores with 600,000+ text-only captions curated by [\[Ghosh et al., 2024\]](#). Third, we will evaluate cross-domain transfer by training on AudioCaps and testing on Clotho (and vice versa) to verify the out-

of-domain generalization claims. If time permits, we will explore replacing the CLAP retrieval mechanism with alternative retrieval strategies, such as BM25 over audio tags, to understand how retrieval quality impacts caption generation.

Resources and Related Work

The dominant paradigm in audio captioning is the encoder-decoder architecture, where a pre-trained audio encoder produces embeddings that condition an autoregressive language decoder [Gontier et al., 2021], [Kim et al., 2023]. [Kim et al., 2023] proposed prefix tuning for audio captioning, feeding a learned prefix embedding sequence to GPT-2, and achieved SOTA results on AudioCaps and Clotho. [Gontier et al., 2021] fine-tuned BART using AudioSet tags as auxiliary conditioning. [Mei et al., 2021] trained a full encoder-decoder transformer from scratch, while [Chen et al., 2020] combined a pre-trained CNN encoder with a transformer decoder. The key innovation that RECAP [Ghosh et al., 2024] introduces over these baselines is the retrieval-augmented generation (RAG) framework, originally proposed by [Lewis et al., 2020] for knowledge-intensive NLP tasks. RAG has been successfully applied to image captioning by [Sarto et al., 2022] and [Ramos et al., 2023], demonstrating that retrieved exemplars can improve caption diversity and accuracy. RECAP adapts RAG to audio by leveraging CLAP [Wu et al., 2023], a contrastive audio-language model that maps audio and text into a shared embedding space, enabling cross-modal retrieval without paired data at inference time.

A notable property of RECAP is its parameter efficiency: only the cross-attention layers (approximately 5.4% of total parameters) are trained, while both the CLAP encoder and GPT-2 decoder remain frozen. This is similar in spirit to adapter-based and prefix-tuning methods that have gained popularity in the broader NLP and vision-language communities.

Datasets

We will primarily use the AudioCaps dataset [Kim et al., 2019], which is the largest publicly available audio captioning benchmark. AudioCaps contains 49,838 training clips, 495 validation clips, and 975 test clips, each approximately 10 seconds long and sourced from AudioSet YouTube videos. Each clip in validation and test sets has five human-written captions, while training clips have one caption each. For cross-domain evaluation, we will also use the Clotho dataset [Drossos et al., 2020], which contains 3,839 training, 1,045 validation, and 1,045 test audio clips from Freesound, each with five captions.

Dataset Links:

- **AudioCaps:** <https://audiocaps.github.io/>
- **Clotho:** <https://zenodo.org/record/3490684>

Evaluation Metrics

We will evaluate our models using the standard audio captioning metrics: BLEU, METEOR, ROUGE-L, CIDEr, SPICE, and SPIDEr (the average of CIDEr and SPICE). These are the same metrics used in the original RECAP paper [\[Ghosh et al., 2024\]](#) and across the audio captioning literature, enabling direct comparison with published results.

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